

Question 1.1. *Let $n \geq 3$ and let V and E be finite-dimensional spaces. Further suppose that the first-order homogeneous linear differential operator with constant coefficients $A(D) : C_c^\infty(\mathbb{R}^n, V) \rightarrow C_c^\infty(\mathbb{R}^n, E)$ is elliptic and canceling. Can one show the existence of a constant $C > 0$ such that the inequality*

$$\|u\|_{L^{\frac{n}{n-1},1}(\mathbb{R}^n, V)} \leq C \|A(D)u\|_{L^1(\mathbb{R}^n, E)}$$

holds for every $u \in C_c^\infty(\mathbb{R}^n, V)$?

Another open question stems from the fact that higher-order operators satisfy Sobolev estimates [41, Theorem 1.3], Hardy inequalities [8, Theorem 1.2] and non-optimal Lorentz estimates [41, Theorem 8.5].

Question 1.2. *Let $n \geq 2$, $k \geq 2$ and let V and E be finite-dimensional spaces. Further suppose that the k -th order homogeneous linear differential operator with constant coefficients $A(D) : C_c^\infty(\mathbb{R}^n, V) \rightarrow C_c^\infty(\mathbb{R}^n, E)$ is elliptic and canceling. Can one show the existence of a constant $C > 0$ such that the inequality*

$$\|D^{k-1}u\|_{L^{\frac{n}{n-1},1}(\mathbb{R}^n, V)} \leq C \|A(D)u\|_{L^1(\mathbb{R}^n, E)}$$

holds for every $u \in C_c^\infty(\mathbb{R}^n, V)$?

The plan of the paper is as follows. In Section 2, we recall some requisite preliminaries regarding Lorentz spaces. In Section 3, we give a proof of Theorem 1 in the plane, as it illustrates well the idea of the more general Theorem 2. The main ingredient for higher dimensions is a version of an inequality of Loomis and Whitney [26] to nonorthogonal coordinate systems. We show how this can be obtained from a change of coordinates and a Lemma of Gagliardo in [17] in a presentation that tries to keep the geometric content of the inequality. In Section 5.1 we prove Theorem 2, from which Theorem 1 is deduced in Section 5.2. Finally, in Section 5.3 we prove a general result for elliptic and $(n-1)$ -canceling operators.

2. PRELIMINARIES

In the sequel, with an abuse of notation we utilize $|\cdot|$ to denote the norm in any finite-dimensional vector space, e.g. the absolute value, the norm in Euclidean space $\mathbb{R}^n$, and the norm in V and E .

For $n \in \mathbb{N}$, $1 < p < +\infty$ and $1 \leq q \leq +\infty$, we denote by $L^{p,q}(\mathbb{R}^n)$ the Lorentz space [27] (see also for example [19]) with quasinorm

$$(2.1) \quad \|u\|_{L^{p,q}(\mathbb{R}^n)}^q \triangleq p \int_0^{+\infty} (t \mathcal{L}^n(\{x \in \mathbb{R}^n : |u(x)| > t\})^{1/p})^q \frac{dt}{t},$$

where $\mathcal{L}^n(A)$ denotes the Lebesgue measure of a measurable set $A \subset \mathbb{R}^n$. Equivalently, if the function $|u|^* : (0, +\infty) \rightarrow [0, +\infty)$ is the nonincreasing rearrangement of $|u|$, that is, if for every $t \in (0, +\infty)$, $\mathcal{L}^1(\{s \in [0, +\infty) : |u|^*(s) > t\}) = \mathcal{L}^n(\{x \in \mathbb{R}^n : |u(x)| > t\})$, then

$$\|u\|_{L^{p,q}(\mathbb{R}^n)}^q = p \int_0^{+\infty} (t \mathcal{L}^1(\{s \in (0, +\infty) : |u|^*(s) > t\})^{1/p})^q \frac{dt}{t} = \int_0^{+\infty} |u|^*(s) s^{\frac{q}{p}} \frac{ds}{s}.$$